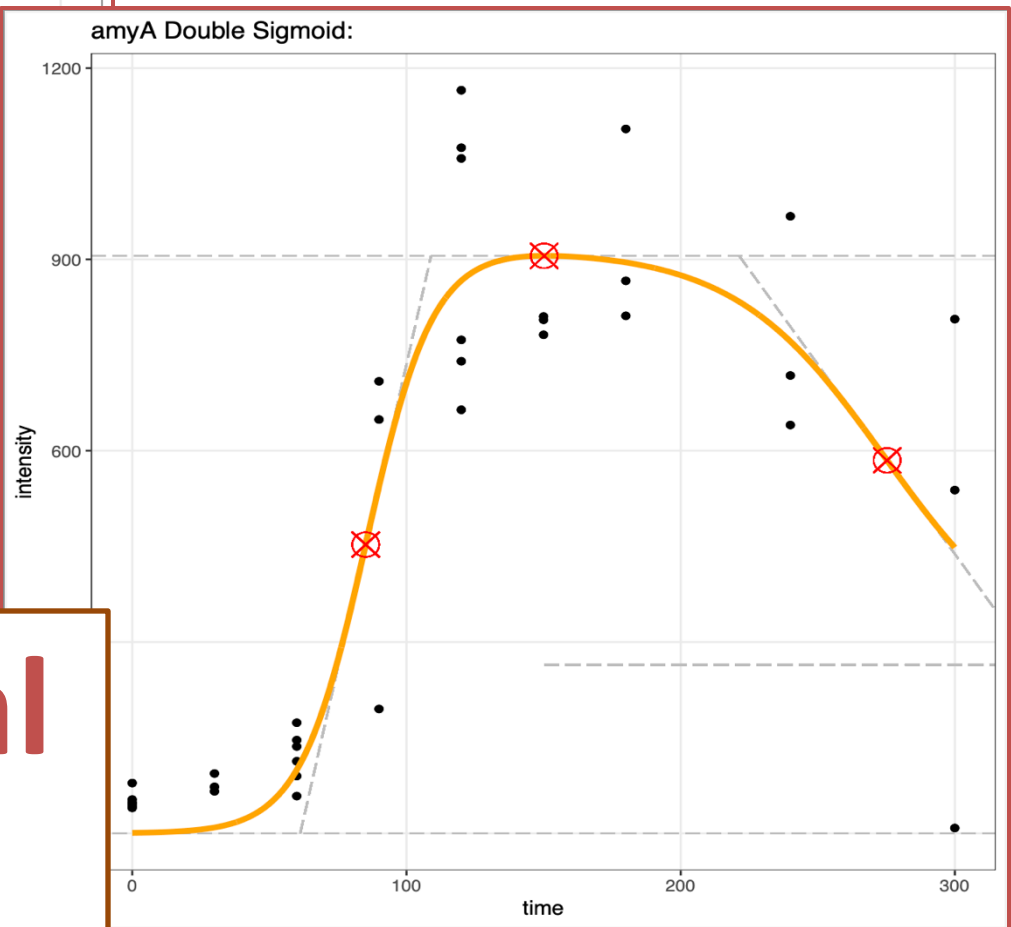
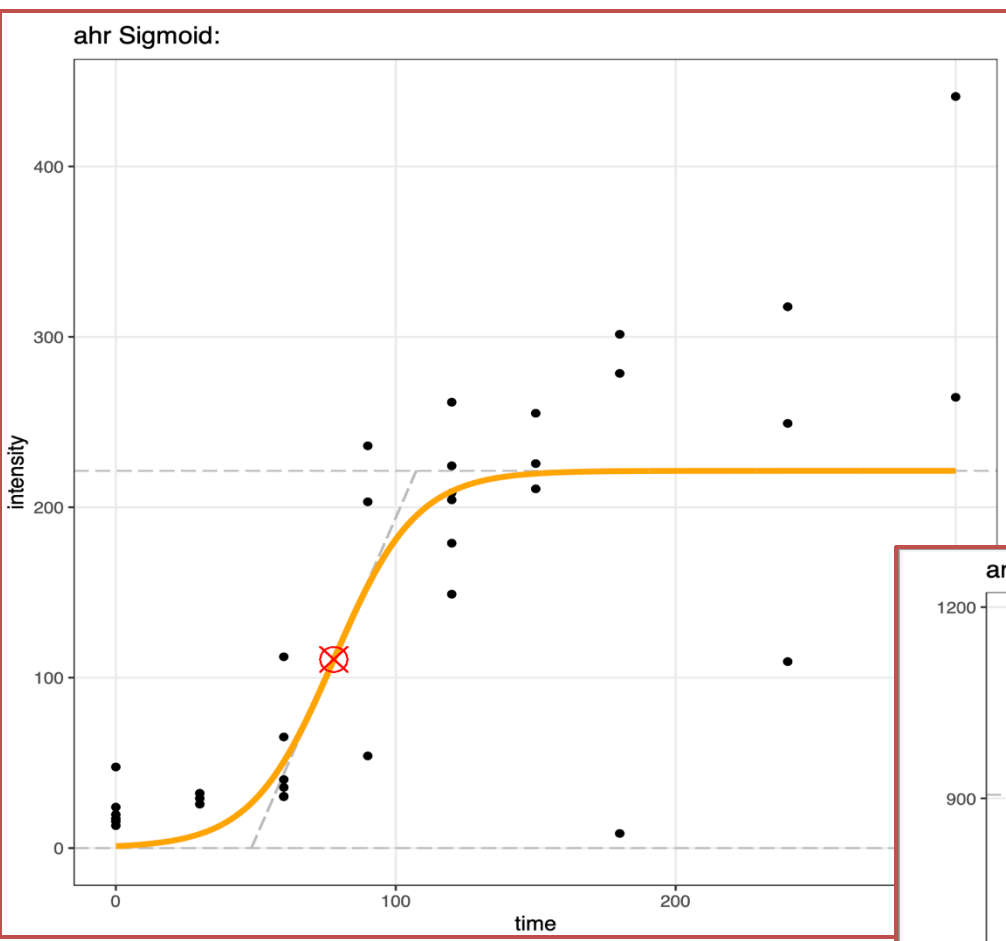
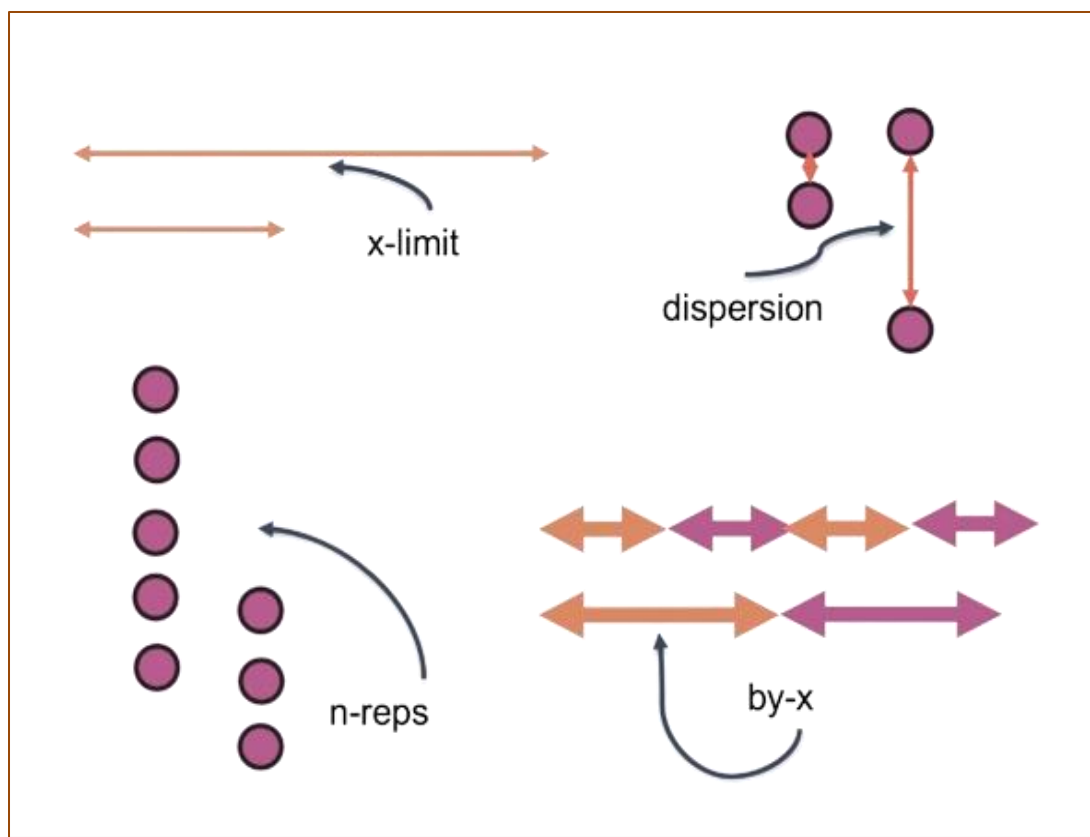
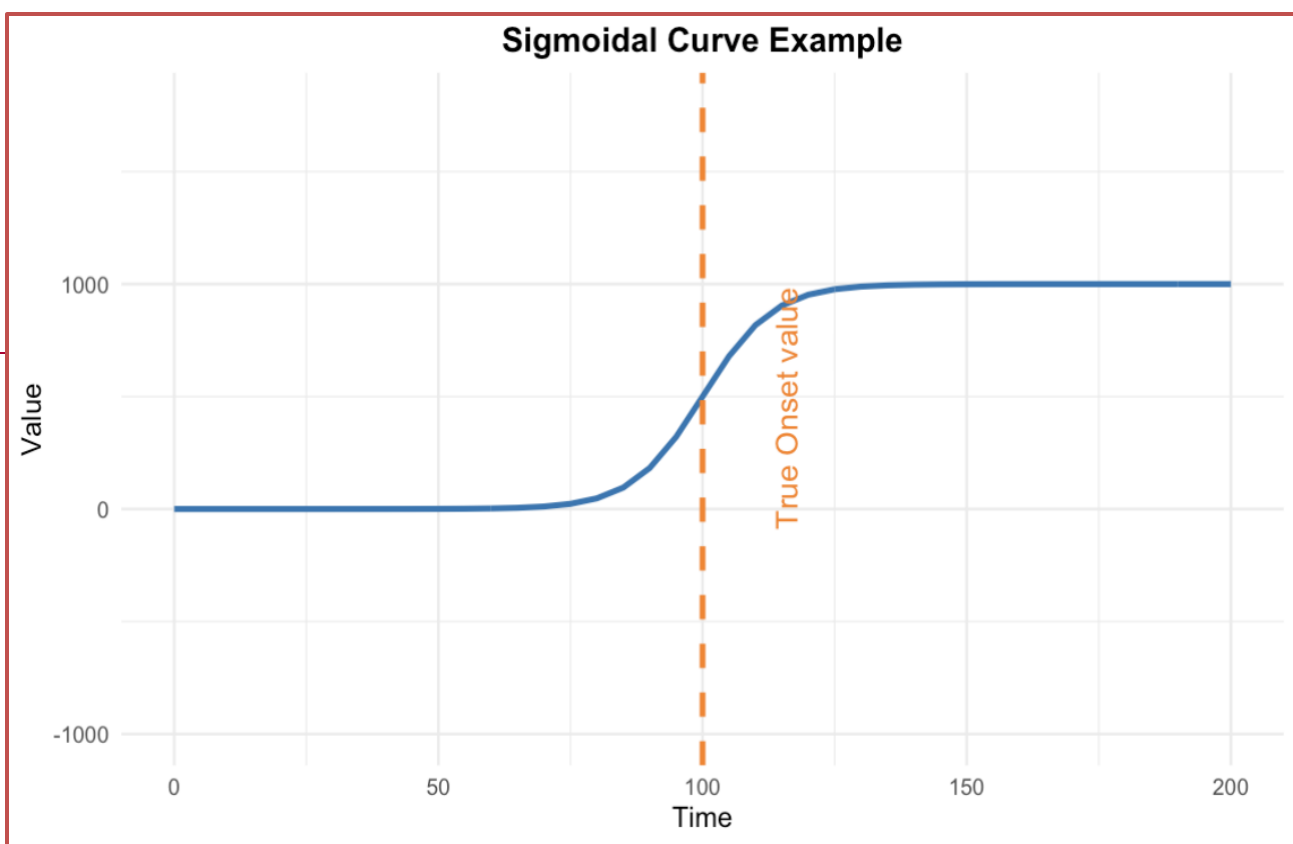


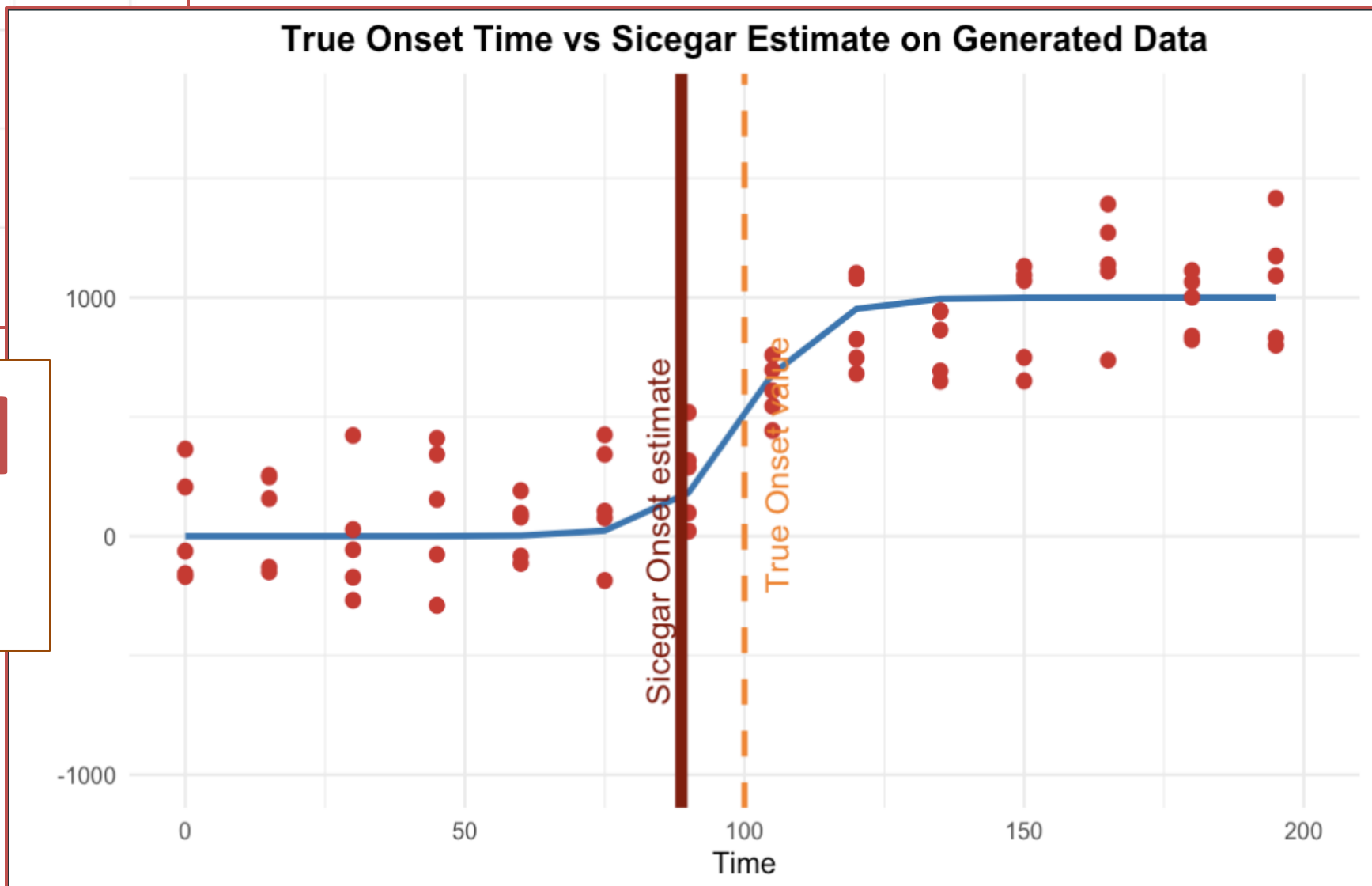
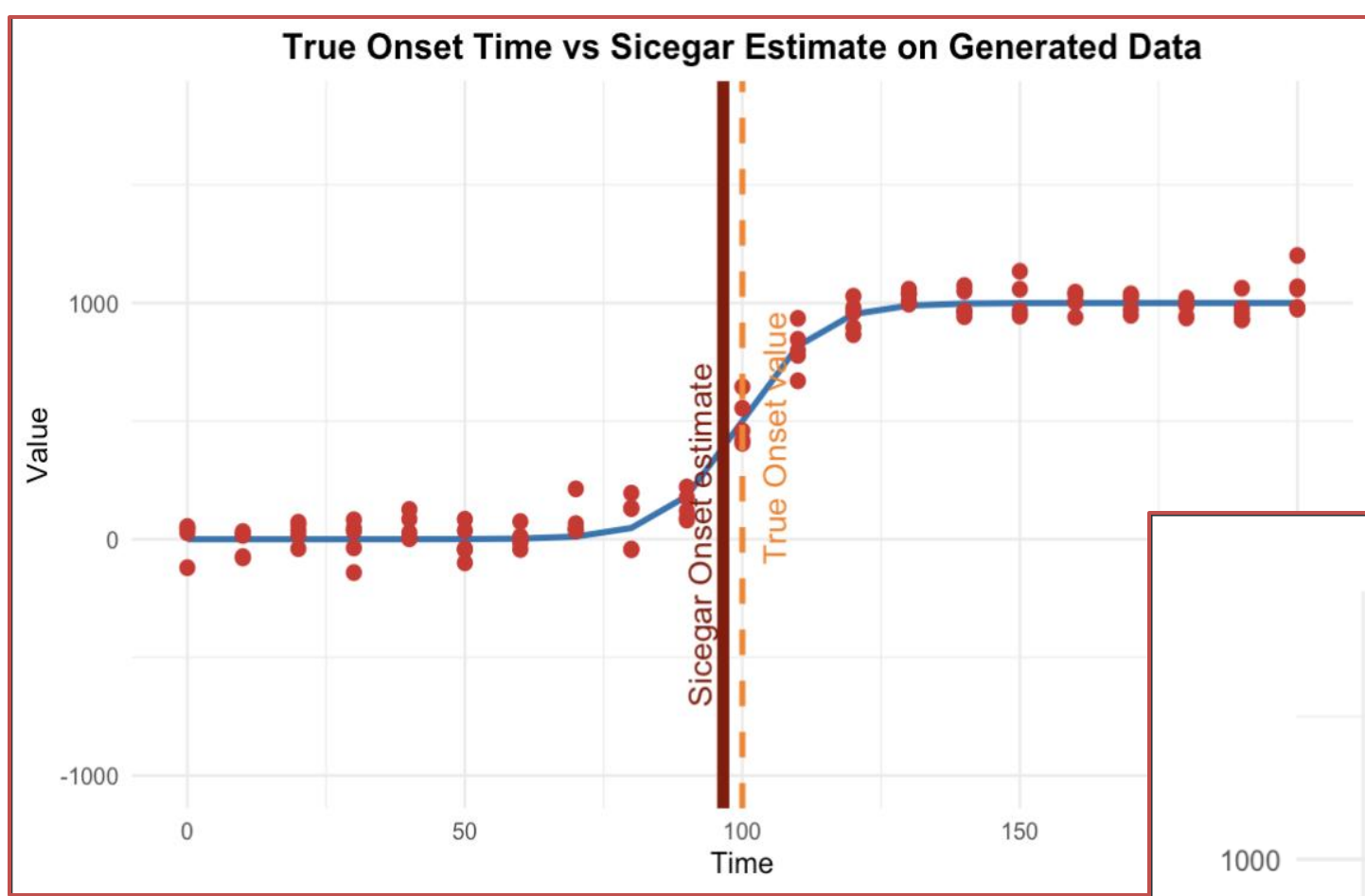
Evaluating the Reliability of Sicegar for Timing Analysis of RpOS-Dependent Gene Expression



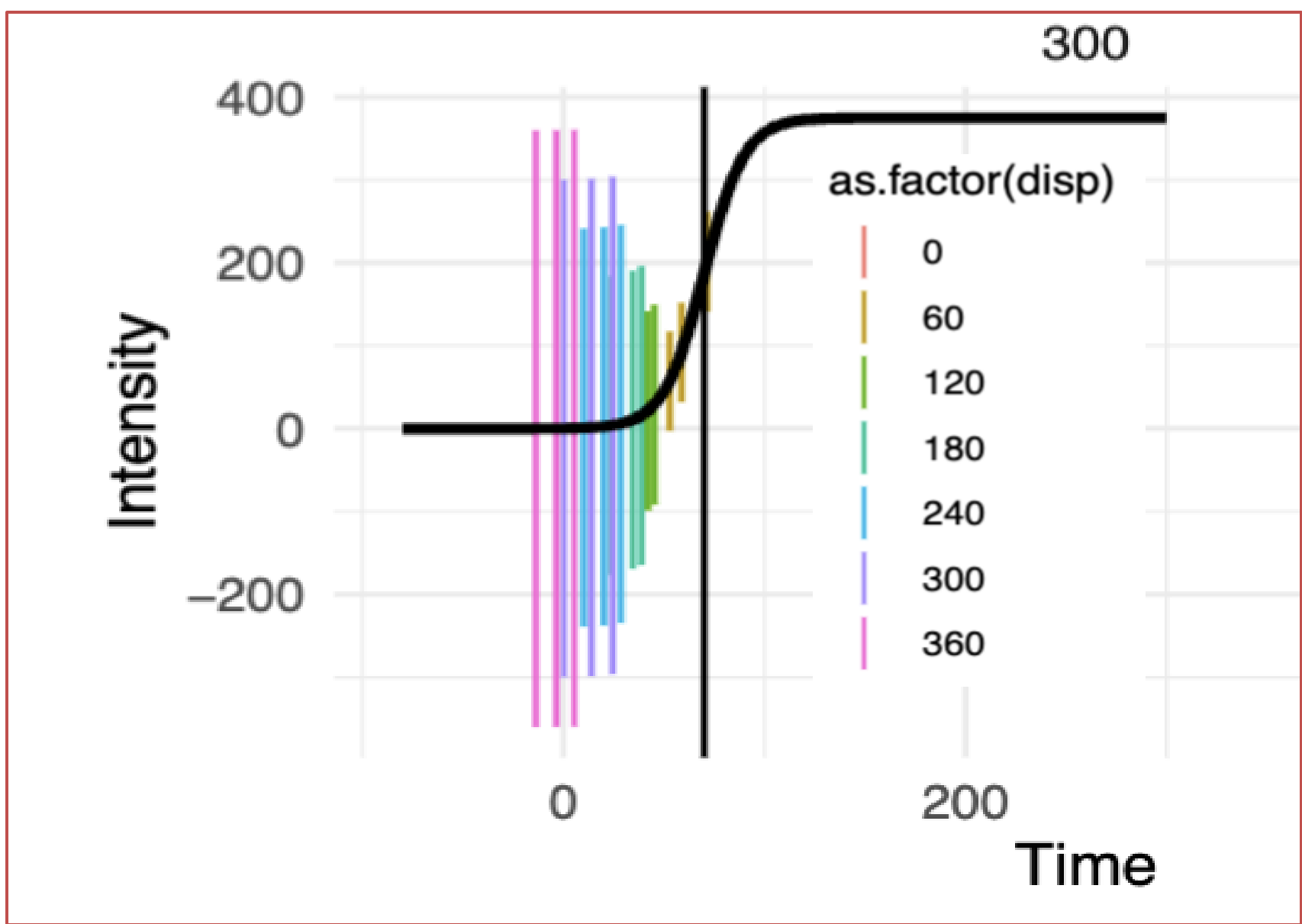
Underlying Model
+
Sampling Decisions



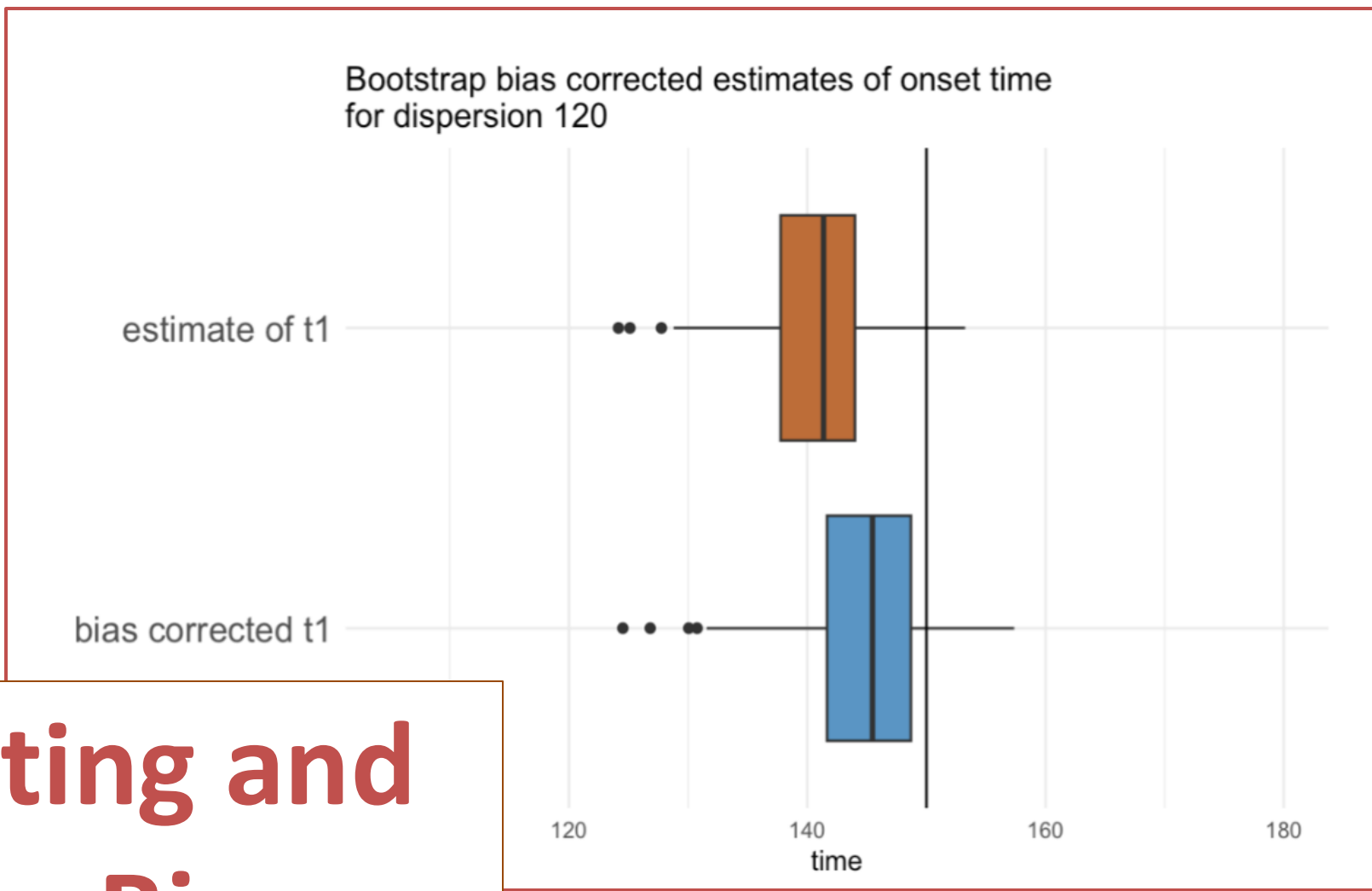
The Original Data



Generating Data and Running Sicegar



Output



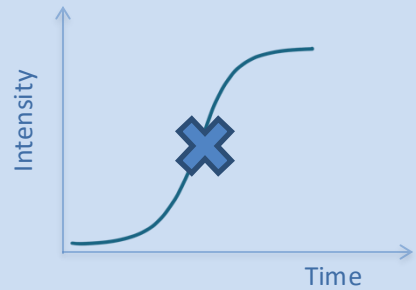
Estimating and Fixing Bias

Introduction & Objectives

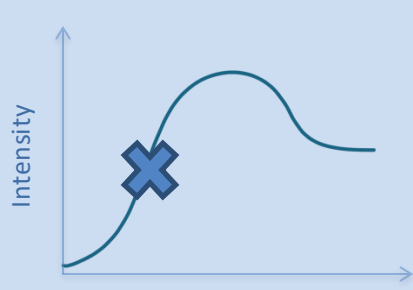
- Gene transcription timing is crucial for understanding cellular stress responses.
- This study evaluates the R package, `sicegar`, ability to model gene expression and accurately estimate transcription onset times of this sort of data.

Methodology

Sicegar Overview: Sicegar fits and classifies gene expression following sigmoidal or double-sigmoidal trends.



$$f(t) = \frac{I_{max}}{1 + e^{-a_1(t-t_1)}}$$



$$f(t) = \frac{1}{1 + e^{-a_1(t-t_1)}} \cdot \frac{1}{1 + e^{-a_2(t-t_2)}}$$

Sigmoidal models show a single-phase S-curve; double-sigmoidal models capture two distinct change phases.

Simulation Design: Synthetic data mimicked real conditions to test `sicegar`'s accuracy, focusing on data dispersion and observation.

Results

Single Sigmoid Data Analysis: Increased data dispersion leads to underestimating onset times, and short data recording duration may cause overestimation.

Double Sigmoid Analysis: The software often underestimates the first onset time but avoids misclassification, leaning towards ambiguity under high variability.

"Ambiguous" Data: High data variability increases the likelihood of ambiguous classifications, reflecting the software's caution.

Bootstrapping to Estimate and Fix Bias

- Bias Correction:** Bootstrapping improved onset time accuracy, reducing systematic bias in high-variability scenarios.
- Single Sigmoid Results:** Bootstrapping effectively aligned onset estimates with true values.
- Double Sigmoid Approach:** Bootstrapping shows potential but is complex and not fully developed for double-sigmoidal models.

References

Adams et al., 2023: Study on gene transcription timing in *E. coli*.
Caglar et al., 2018: `sicegar` package documentation.
Efron, 1993: Bootstrapping methods.

Acknowledgements

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